

ABDIKARIN MOHAMED

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EDUCATION

University of Minnesota – Twin Cities	Dec 2024
Bachelor of Science in Computer Science	
The University of Texas at Austin	Dec 2027
Master of Science in AI and Machine Learning	
Relevant Coursework: Advanced Deep Generative Models, Online Learning and Optimization	

EXPERIENCE

American Express	Jan 2025 – Present
<i>Software Engineer</i>	<i>Phoenix, AZ</i>
- Engineered scalable backend services for the Amex Offers Partner Layer (AOPL), a high-throughput API gateway enabling external partners (Rakuten, CDLX, Krowd) to integrate with internal systems. - Designed and implemented a rule-based orchestration engine using Java and Cassandra to transform, validate, and route partner payloads, reducing onboarding time by 40%. - Built and optimized data processing pipelines handling 500K+ monthly payloads, including validation, enrichment, and response filtering, improving reliability and data accuracy by 30%.	
American Express	Jun 2024 – Aug 2024
<i>Software Engineering Intern</i>	<i>Phoenix, AZ</i>
- Developed backend systems for high-volume transaction ingestion pipelines, processing merchant payment data at scale. - Built a data ingestion and transformation pipeline using Apache Camel to improve reliability and throughput of transaction processing. - Implemented secure data handling and preprocessing utilities for sensitive financial data, ensuring compliance and integrity.	
Microsoft	Jun 2023 – Aug 2023
<i>Software Engineer Intern</i>	<i>Redmond, WA</i>
- Built an AI-assisted communication tool leveraging text-to-speech and NLP techniques to support users with speech impairments. - Developed a responsive frontend using React, TypeScript, Redux, and Material-UI, integrating AI-driven features for real-time interaction. - Presented the product to Microsoft VP leadership, demonstrating usability and measurable user impact.	

PROJECT

Retrieval-Augmented Generation (RAG) System – In Progress
- Built an end-to-end RAG pipeline using Python, embeddings, and vector search for domain-specific knowledge retrieval. - Implemented chunking strategies, semantic retrieval, and prompt orchestration to improve response grounding and reduce hallucinations. - Designed evaluation workflows using structured test queries to measure retrieval precision and response quality.

TECHNICAL SKILLS

- Languages:** Python, Java, TypeScript, SQL
- AI & LLMs:** OpenAI API, Azure OpenAI, RAG pipelines, embeddings, prompt engineering
- Data Engineering:** ETL pipelines, distributed systems, data transformation, structured/unstructured data processing